

Class #14 - Inverse of a block matrix

1. Def: Let u and v be vectors in $\mathbf{R}^n$.

The product $u^T v$ is called the *scalar product* or *inner product* of u and v (It is usually written as a single real number without brackets).

The product uv^T is called the *outer product* of u and v .

2. Example: Let $A = \begin{bmatrix} -3 & 1 & 2 \\ 1 & -4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$. Verify that

$$AB = \sum_{k=1}^3 \text{col}_k(A) \text{row}_k(B).$$

3. Column-Row expansion of AB :

If $A \in \mathbf{R}^{m \times n}$ and $B \in \mathbf{R}^{n \times p}$, then

$$AB = \begin{bmatrix} \text{col}_1(A) & \text{col}_2(A) & \cdots & \text{col}_n(A) \end{bmatrix} \begin{bmatrix} \text{row}_1(B) \\ \text{row}_2(B) \\ \vdots \\ \text{row}_n(B) \end{bmatrix} = \sum_{k=1}^n \text{col}_k(A) \text{row}_k(B).$$

Proof: $(AB)_{ij} = \underline{\hspace{2cm}}$:

Note from $\text{col}_k(A) \text{row}_k(B)$, we get:

Hence, $(AB)_{ij} = \underline{\hspace{2cm}}$ which is the (i, j) -entry in AB , by the row-column rule.

4. Example: Let

$$A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}.$$

Such A and B are called *partitioned matrices* or *block matrices*. When the partitions of A and B are conformable for block multiplication, then the ordinary product AB can be written as

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix}.$$

5. Example: A matrix of the form

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$

is said to be *block upper triangular*. Suppose A_{11} is $p \times p$, A_{22} is $q \times q$, and A is invertible. Find a formula for A^{-1} .

Denote A^{-1} by B and partition B :

We get 4 equations:

So, $A^{-1} =$